Structural and Material Integration of a SERF MEMS Magnetometer Payload in an Autonomous Underwater Vehicle

Towed-Body Mechanics, Magnetic Cleanliness, and a Geometry-Derived Rejection Limit

Scientific Report — Companion to the SERF-MAD Feasibility Study

16 August 2026

Provenance convention. Every quantitative claim is marked: [M] measured and cited [L] literature closed form evaluated here [S] simulated [A] engineering assumption. Unmarked numbers are restated from the parent feasibility study.

Executive Summary

The parent feasibility study [1] established that a ruggedised SERF MEMS magnetometer is a physically credible AUV magnetic anomaly detection payload, and that its feasibility is governed not by sensor sensitivity — which exceeds requirement by five orders of magnitude [3, 5] — but by ambient-field control and platform magnetic cleanliness. It derived a residual-field requirement of $\leq 0.94\,nT$ (94 dB of suppression) from the Cross-Axis Projection Error [4], and demonstrated by sensitivity analysis that the system is clutter-limited: the elasticity of detection range with respect to sensor noise is exactly zero, while clutter magnitude, rejection ratio and gradiometer baseline all register near 0.2.

That study deliberately placed hydrodynamic design out of scope, represented passive shielding as a scalar multiplier, and represented platform self-field as a parametric common-mode rejection ratio rather than deriving it from vehicle geometry and materials. This report addresses those three gaps. It executes Recommendation 3 of the parent study — replace the parametric rejection ratio with a geometry-based vehicle magnetic model — and reports three findings.

1. **Gradiometric rejection of the platform is limited by geometry, not by channel matching.** A two-cell gradiometer rejects a *uniform* field to the limit set by gain and axis matching, and for the 0.1% matching of requirement R-09 that is 60 dB. The vehicle's own sources are not uniform: they sit metres from the baseline, so their field has a genuine gradient across it which no amount of matching removes. Deriving the rejection from component placement gives 5.5 dB at a 2 m standoff[S], not 60 dB. Reaching 60 dB against a near-field source by standoff alone would require a 1500 m boom. The parent study anticipates this mechanism qualitatively in its test X-07 discussion; this report quantifies it.
2. **Magnetic cleanliness is dominated by remanence, and remanence is a build-time property.** Induced magnetisation scales with the ambient field and therefore vanishes inside a shield; remanent magnetisation does not. Substituting the material of an otherwise identical fastener set changes the field at the sensor by more than six orders of magnitude, from $999.9\,pT$ for cold-worked 316L to $4.6\times10^{-4}\,pT$ for CFRP[S]. The ranking is robust; the magnitudes rest on assumed remanence values and are flagged accordingly.
3. **The passive shield closes the 94 dB budget only if it is annealed in its final form.** The $0.94\,nT$ requirement implies 53,191× total suppression, hence a 532× passive stage against the study's 100× active stage[L]. An annealed mu-metal cylinder of 5 cm diameter and 1 mm wall delivers 1119×[L]. The same alloy formed from foil and not re-annealed delivers 93× and fails the requirement by a factor of 5.7[L], because forming degrades the measured initial permeability from 55,955 to 4,660 [9]. Re-annealing requires >1000°C and is not available to an assembled payload.

The operational consequence of finding 1 is material. Propagating the derived rejection through the parent study's unmodified Monte Carlo detection chain reduces R_{50} against a naval-mine-class target from

66.75 m to 15.07 m at a 2 m standoff[S]— 23% of the figure obtained under the assumed rejection. Section 7 states what this does and does not imply.

1 Introduction

1.1 Inherited position

Magnetic anomaly detection is the one underwater sensing modality unaffected by acoustic quieting [1]. Spin-exchange relaxation-free magnetometers [2] in microfabricated vapour cells [3] make a picotesla-class sensor small and robust enough for an autonomous underwater vehicle. The parent study modelled the sensor quantum-mechanically, the target electromagnetically, the vehicle kinematically and detection statistically, and its twenty-seven verification, validation and stress tests all pass.

Two of its conclusions are imported here without re-derivation. First, the Cross-Axis Projection Error requirement: a residual transverse field tilts the effective sensitive axis by $\alpha = \arctan(\gamma_e B_{\mathrm{res}}/R_{\mathrm{tot}})$ and scales the on-axis gain by $\cos\alpha$ [4], so a 5% gain loss criterion gives

$$B_{\mathrm{res}} \le \frac{R_{\mathrm{tot}}}{\gamma_e}\tan\!\big(\arccos 0.95\big) = 0.94\,nT, \tag{1}$$

a factor 5.3 tighter than the nominal SERF window and equivalent to 94 dB of combined suppression. Second, the clutter-limited finding: a $5\,fT/\sqrt{\mathrm{Hz}}$ cell [3] sits roughly five orders of magnitude below the measured post-compensation UUV interference residual of $2.23\,nT$ [5], so improving the magnetometer changes nothing while improving vehicle magnetic cleanliness improves range almost proportionally.

1.2 What was left open, and what this report adds

The parent study records three limitations that this report is written to address. Its Section 9.1(2) states that platform self-field is parametric — "represented by a rejection ratio and a $1/f$ residual rather than derived from hull geometry and materials". Its Section 2 places hydrodynamic design out of scope. And its passive shielding enters as a scalar factor with no material basis. Requirement R-13 ("boom or tow displacement maximising distance from thrusters and battery") is consequently stated without a distance, and R-10 ("non-magnetic fasteners") without a material.

Before this work no material property existed anywhere in the simulation package: no susceptibility, no remanence, no conductivity. A material choice therefore could not influence any result, and any "simulation-backed" material claim would have been vacuous. Section 4 describes the property layer built to close that gap, and Section 5 the vehicle model that consumes it. Verification test M-08 exists precisely to assert that the path is live: it fails if substituting a material ever stops changing a computed field.

2 Method and Provenance Discipline

The parent study holds physical constants in one module with inline citations and stores modelling assumptions in a distinct data type so they cannot be mistaken for measured values in any downstream table. That discipline is extended rather than relaxed. Every property in the new material schema is wrapped in exactly one of two types — a citation-carrying `Measured` or the pre-existing `EngineeringAssumption` — and verification test M-11 asserts that no bare number can enter the schema.

The result is uncomfortable and is reported as such: 84.0% **of the property values in the schema are engineering assumptions rather than measurements**[S]. This is not a defect of the schema but an accurate statement about the open literature on the magnetic properties of engineering alloys. Handbook susceptibilities are quoted for annealed, contaminant-free material; a real fastener has been cold-headed and thread-rolled, and no measured remanence for such a part was located. Section 8 treats the consequences.

All new work is strictly opt-in. Verification test M-10 asserts that the pre-existing model defaults are untouched, and the parent study's suites were re-run after every change: 17/17 verification and validation tests pass, 10/10 stress tests pass, and all eleven published figures remain byte-identical. The materials suite adds 11/11 tests in a separate module so the published counts do not move. GPU results are gated on the verification kernel, which agrees with the closed-form Bloch fixed point to 9.99×10^{-15} against a 10^{-10} tolerance with zero inter-replica spread.

3 Hydrodynamic Stabilisation and Towed-Body Mechanics

3.1 Scope of the evidence

This section is **literature only**. No hydrodynamic solver exists in the simulation package and building one was out of scope; no drag, lift or vortex-shedding quantity in this section is simulated. Where a measured figure exists it is cited and marked [M]; where the open literature does not supply one, that is stated rather than estimated. Two consequences *are* computed, using the parent study's already-validated attitude and field-budget code, and are marked [S].

3.2 Rigid mounting versus tow cable

The strongest open-literature evidence favours a rigid mount over a tow cable for a gradiometer. The GEOMAR Girona 500 hovering AUV carries four SENSYS FGM3D/100 UW II three-axis fluxgates sampled at 200 Hz, giving 2.5 mm along-track spacing at 0.5 m/s, in a geometry that resolves all three spatial magnetic gradients [7][M]. That system states its design rationale directly: the rigid construction holding the magnetometers suppresses spatial errors that towed systems usually produce [7]. The sensors sit 2 m from the vehicle nose and approximately 3 m from the rear thrusters, which are identified as the principal source of carrier noise [6][M].

The mechanism disfavouring a cable is well documented. Vortex shedding excites transverse cable vibration — strum — and when the shedding frequency approaches a cable natural frequency, lock-in produces large displacements that are transmitted into the towed element. Passive suppression by fairings, helically distributed stubs, or float–sinker distributions is standard practice. For a *gradiometer* the objection is sharper than for a single sensor: requirement R-09 asks that a ≥ 0.5 m baseline hold gain and axis matching to $\leq 0.1\%$, and a compliant member that permits relative motion between the two cells corrupts the baseline itself, which enters the measurement directly.

Gap. No measured comparison of gradiometer baseline stability between a rigid boom and a tow cable was located. The GEOMAR statement is qualitative. This is a bench-testable question and appears in Section 9 as a gating experiment.

3.3 Measured towed-body attitude, and its consequence

Where towed configurations are used, measured attitude figures do exist. A twin-body towed magnetometry system using an S-shaped cable to decouple platform motion reports a mean cruising speed of 0.56 m/s with the towed body's pitch held within $\pm 5°$ and roll within $\pm 1°$, achieving $0.0639\,nT$ dynamic noise and repeat-line and cross-line internal consistency of $1.832\,nT$ and $1.956\,nT$ [8][M].

That pitch figure deserves attention, because it is *worse* than the attitude the parent study assumes for a self-propelled hull in a moderate sea (3° roll, 1.5° pitch). Towing is not automatically stabilising. Propagating the measured towed-body attitude through the parent study's validated attitude and nulling model gives Table 1.

Attitude case	Motion swing	Residual	CAPE tilt	Gain
Modelled moderate sea, 3.0°/1.5°	2,761.6 nT	4.742 nT	58.8°	0.518
Measured towed body, 1°/5° [8]	4,128.1 nT	4.831 nT	59.3°	0.511

Table 1: Attitude-induced body-frame field swing and its CAPE consequence[S], computed with the parent study's unmodified `physics.platform` module. The first row reproduces the parent study's published field budget exactly, which confirms correct use of that model.

The measured towed-body attitude produces a 1.49× larger motion-induced swing[S]. The residual after nulling barely moves, because at the baseline 100× passive and 100× active configuration the residual is dominated by the static field rather than by the swing, so the CAPE penalty is essentially unchanged. The consequence falls instead on the *servo*: the peak slew implied by a $4{,}128\,nT$ swing at 0.40 Hz is approximately $5{,}200\,nT/s$[S], which remains inside the $\geq 8{,}000\,nT/s$ of requirement R-03. R-03 therefore covers the measured towed case, and that is worth knowing explicitly rather than assuming.

3.4 Standoff under r^{-3} decay

Vehicle sources fall as r^{-3} [20], so displacement is the cheapest available reduction in absolute clutter: doubling standoff reduces the common-mode field eightfold. Section 5 shows that the same displacement does *not* buy a proportionate improvement in gradiometric rejection, and that the two effects must not be conflated.

Gap. No measured magnetic signature of a UUV thruster as a function of distance was located in the open literature. The defensible whole-vehicle anchor remains the $246.90\,nT$ raw interference measured before compensation [5], and it is used as the calibration point in Section 5. No NACA-profile fin data specific to magnetometer towed bodies was located either.

4 Material Selection and Magnetic Cleanliness

4.1 Susceptibility and remanence are different problems

A component's dipole moment has two parts. The induced moment $\mathbf{m}_{\text{ind}} = \chi V \mathbf{H}$ follows the ambient field and is therefore proportional to it; the remanent moment $\mathbf{m}_{\text{rem}} = M_r V \hat{\mathbf{d}}$ has its own fixed body-frame direction and is independent of it. Verification test M-04 asserts this separation numerically: the induced term doubles when the ambient field doubles and stays parallel to it, while the remanent term is exactly constant[S].

The engineering consequence is decisive and is the reason this is a materials report rather than a processing one. **Nulling the ambient field removes the induced moment and leaves the remanent one.** Magnetic hygiene is therefore a build-time constraint that cannot be recovered later by field control, by compensation, or by gradiometry.

4.2 The cold-work trap

Austenitic stainless steels are specified as non-magnetic and are so when annealed: measured DC permeabilities for annealed 316/317L are 1.003–1.005 at 200 oersteds [14][M]. Cold work transforms austenite to strain-induced martensite, which is ferromagnetic, and raises permeability [15]. A fastener is precisely a cold-worked part — headed and thread-rolled — so a bolt certified to a non-magnetic grade need not behave like the annealed bar it was made from.

Grades with high austenite stability resist this. Nitrogen-strengthened Nitronic 50 (XM-19) retains a typical annealed permeability of 1.004 and remains low-permeability in the cold-worked condition and at cryogenic temperature [16, 15][M].

Table 2 substitutes an otherwise identical fastener set through the candidate materials. Only the material changes; volume, position and geometry are held fixed, so the entire spread is attributable to material choice.

Material	χ (SI)	M_r (A/m)	Field at sensor	Provenance of χ
316L, cold worked	5×10^{-1}	500	$999.9\,pT$	assumed [A]
316L, annealed	4×10^{-3}	5	$9.971\,pT$	measured [14] [M]
Nitronic 50 (XM-19)	4×10^{-3}	2	$4.055\,pT$	measured [16] [M]
Ni-Al bronze (C63000)	1×10^{-4}	1	$1.975\,pT$	assumed [A]
Ti-6Al-4V (Grade 5)	1.8×10^{-4}	0.5	$0.9911\,pT$	assumed [A]
CFRP, quasi-isotropic	-1×10^{-5}	0	$4.6 \times 10^{-4}\,pT$	assumed [A]

Table 2: Fastener material substitution at fixed geometry[S]. **Every M_r in this table is an engineering assumption**: no measured remanence for a worked fastener of any of these alloys was located. The ranking is robust because it follows from the presence or absence of a ferromagnetic phase; the magnitudes are not.

The recommendation that follows is narrow and actionable: specify **Nitronic 50 for fasteners** rather than 316L, on the documented grounds that it does not form martensite on cold work [15, 16], and prefer CFRP for non-load-bearing payload structure.

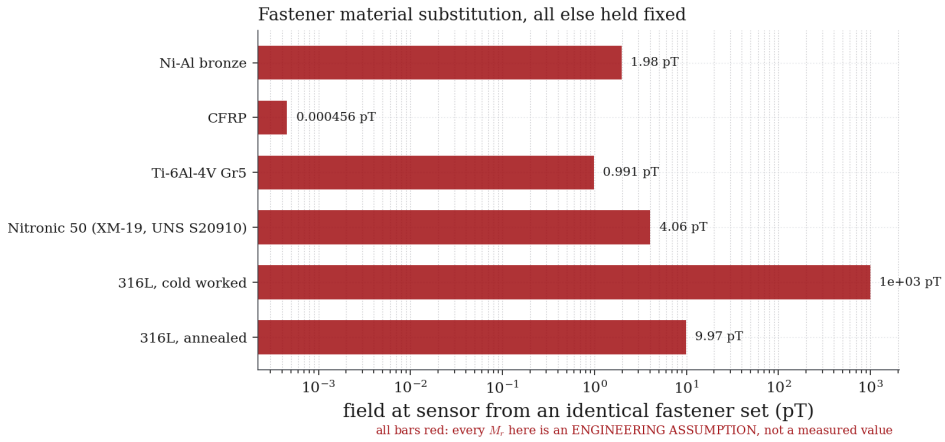


Figure 1: Field at the sensor from an identical fastener set in each candidate material[S]. All bars are drawn in the assumption colour because the remanence driving each is assumed, not measured.

4.3 Shielding materials and the forming penalty

Requirement R-02 asks for ≥ 94 dB of combined suppression. Equation (1) implies 53,191× total; against the parent study's 100× active stage the passive stage must therefore supply 532×[L].

For a single-layer cylindrical shield the amplitude response is $T = D/(\mu_r \Delta)$ for diameter D and wall thickness Δ [10], so the shielding factor is

$$S = \frac{\mu_r \, \Delta}{D}. \tag{2}$$

This is a literature closed form; the fuller cylindrical treatment is [12] and the multi-shell case, which is *not* modelled here, is [11, 13]. Evaluating Eq. (2) with measured permeabilities [9] gives Table 3.

Condition	μ_r	D	Δ	S	**Meets 532×?**
Annealed in final form	55,955 [M]	5 cm	1.0 mm	1119×	yes (2.1× margin)
Annealed in final form	55,955 [M]	10 cm	1.0 mm	559×	marginal (1.05×)
Foil, formed, not re-annealed	4,660 [M]	5 cm	1.0 mm	93×	**no** (short 5.7×)
Foil, formed, not re-annealed	4,660 [M]	10 cm	1.0 mm	47×	**no** (short 11.4×)

Table 3: Passive shielding from Eq. (2)[L], using the initial permeabilities measured in [9]. Permeabilities are measured; the shielding factors are a closed form evaluated here, not a simulation result.

Two conclusions follow, and the second is the more important.

First, the budget *closes* — but not with margin to spare. An annealed mu-metal shield of 5 cm diameter and 1 mm wall clears the requirement by a factor of 2.1. At 10 cm, a more plausible envelope once coils, oven and insulation are accommodated, the margin is 1.05 and effectively gone. Shield diameter is thus a first-order payload architecture parameter, not a detail.

Second, **the same alloy fails if it is formed and not re-annealed**. [9] measured $\mu_i = 55{,}955 \pm 7$ for an annealed cylinder but only 3,670, 3,602 and 4,660 for three shields rolled from foil and not re-annealed, attributing the loss to deformation during forming[M]. Recovery requires annealing above 1000°C, which cannot be applied to an assembled payload. For a vehicle-mounted instrument that must survive launch, recovery, vibration and pressure cycling this is a live risk: a shield deformed in service does not fail loudly, it quietly stops shielding, and the parent study notes that loss of SERF lock does not degrade gracefully.

4.4 Thermal materials

Requirement R-07 asks for 150–200°C at ±5 mK using non-magnetic heating [19]. Silica aerogel is the preferred insulation on two independent grounds: its thermal conductivity is roughly half that of still air,

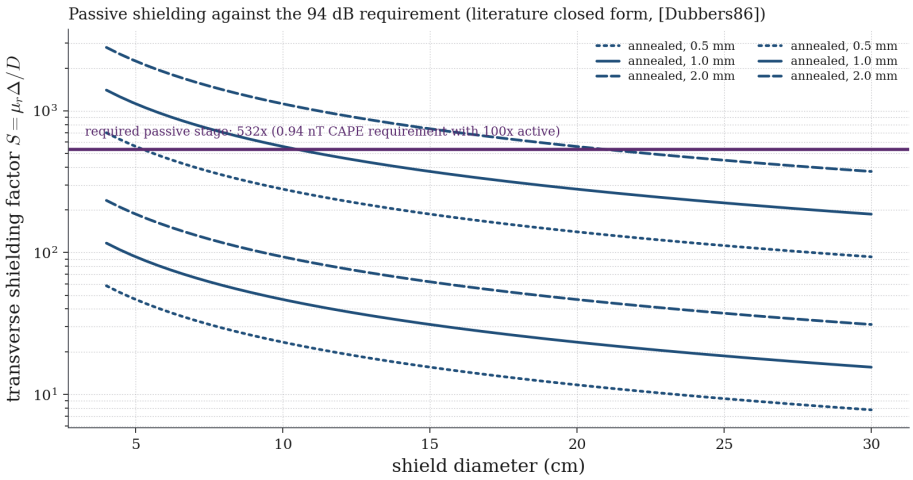


Figure 2: Shielding factor against shield diameter for annealed and formed mu-metal at three wall thicknesses[L], with the 532× passive requirement marked.

which is what makes the budget tractable in cold seawater, and it is an electrical insulator, so unlike a metallic jacket it can support neither eddy currents nor Johnson noise near the cell [18]. Both figures are assumptions in the present schema[A].

Two incompatibilities are worth recording. The epoxy matrix of CFRP has a service limit at the *bottom* of the cell operating range, so CFRP must not contact the cell oven[A]. And nickel-aluminium bronze, the traditional non-magnetic marine fitting alloy, has the highest conductivity of any candidate here (4.3×10^6 S/m[A]) and is therefore the material most likely to become a Johnson-noise problem adjacent to the cell — a term this work does not yet compute, as Section 8 states.

5 Integration Architecture: A Geometry-Derived Rejection Limit

5.1 Two rejection regimes

A two-cell gradiometer forms $(B_1 - B_2)/L$. How well it rejects an interferer depends on where that interferer is, and the two regimes differ by orders of magnitude.

For a **distant, uniform** source the field is equal at both cells to high accuracy, so the differential output is set purely by how well the channels are matched. With the $\leq 0.1\%$ gain and axis matching of R-09 the rejection is $1/10^{-3} = 60$ dB. *This is the regime the widely quoted figure describes, it applies to the Earth's field, and nothing in this report disputes it.*

For a **near-field** source — a thruster or battery metres from the baseline — the field genuinely differs between the cells. That difference is a real differential signal, not a mismatch artefact, and no amount of channel matching removes it. For a dipole at standoff d with baseline L along the same axis, $B \propto r^{-3}$ gives a differential-to-common ratio of approximately $3L/d$, hence

$$\mathrm{CMRR} \approx \frac{d}{3L}. \tag{3}$$

5.2 Model and calibration

The vehicle is represented as a superposition of point-dipole sources at body-frame positions, evaluated with the parent study's already-verified dipole and gradient-tensor code (its tests V-01 to V-03 pass at 10^{-12}, 7.4×10^{-10} and 6.4×10^{-16}). Structural components take their moments from the material schema; motors and battery packs instead carry an explicit assumed moment, because a brushless motor's magnetic circuit closes its flux through the stator so its net external moment is far below the sum of its magnets', and a battery's moment is dominated by DC current loops rather than by material susceptibility. Deriving

those from $M_r V$ would be a fabricated calculation, and the schema records which route each component used.

The reference REMUS-class layout is calibrated so that its total field at the baseline centre reproduces the 246.90 nT raw UUV interference measured before compensation [5] — the only measured anchor available. Verification test M-07 asserts this to a relative error below 10^{-9}[S]. The calibration constrains the *total* only; the component split remains an assumption.

5.3 Result

Table 4 and Figure 3 give the derived rejection. Verification test M-02 confirms the full vector superposition converges to Eq. (3) in the far field, and M-03 that rejection increases monotonically with standoff, which is the quantitative content of R-13.

Standoff	Derived CMRR	Shortfall vs 60 dB	R_{50}	Binding mechanism
1 m	1.9 dB	58.1 dB	13.89 m	geometry
2 m	5.5 dB	54.5 dB	15.07 m	geometry
3 m	8.1 dB	51.9 dB	15.88 m	geometry
5 m	11.7 dB	48.3 dB	17.16 m	geometry
10 m	17.1 dB	42.9 dB	19.90 m	geometry
assumed	60.0 dB	—	66.75 m	matching

Table 4: Derived common-mode rejection and the resulting detection range against a naval-mine-class target[S], baseline 0.5 m. Geometry binds at every realistic standoff. The assumed row reproduces the parent study's published R_{50} of 67.6 m to within its documented seed spread.

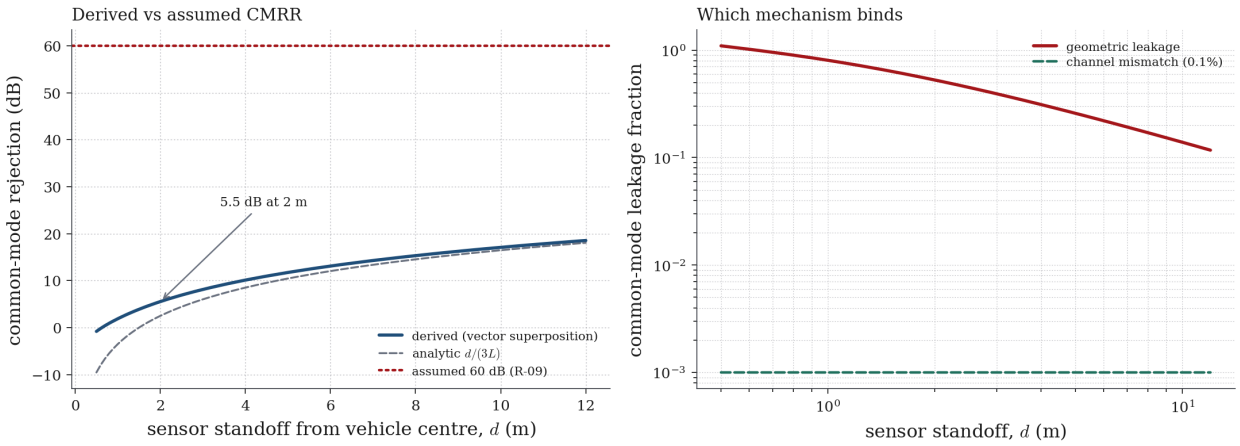


Figure 3: Left: derived rejection against standoff[S], with the analytic $d/(3L)$ estimate and the assumed 60 dB. Right: the two leakage mechanisms. Geometric leakage exceeds channel mismatch by two to three orders of magnitude throughout the plausible range.

5.4 Detection consequence

The derived rejection was propagated through the parent study's *unmodified* Monte Carlo detection chain, with identical range bounds, trial count and seed; the only argument changed between rows is the rejection ratio, so the difference is attributable to it alone. Detection range falls to 21–30% of the figure obtained under the assumed rejection (Figure 4).

5.5 Why standoff cannot fix this

Equation (3) is linear in d while the absolute field falls as d^{-3}, and the difference matters. Reaching 60 dB of *rejection* by displacement alone would require $d = 3L/10^{-3} = 1500$ m at a 0.5 m baseline. A boom

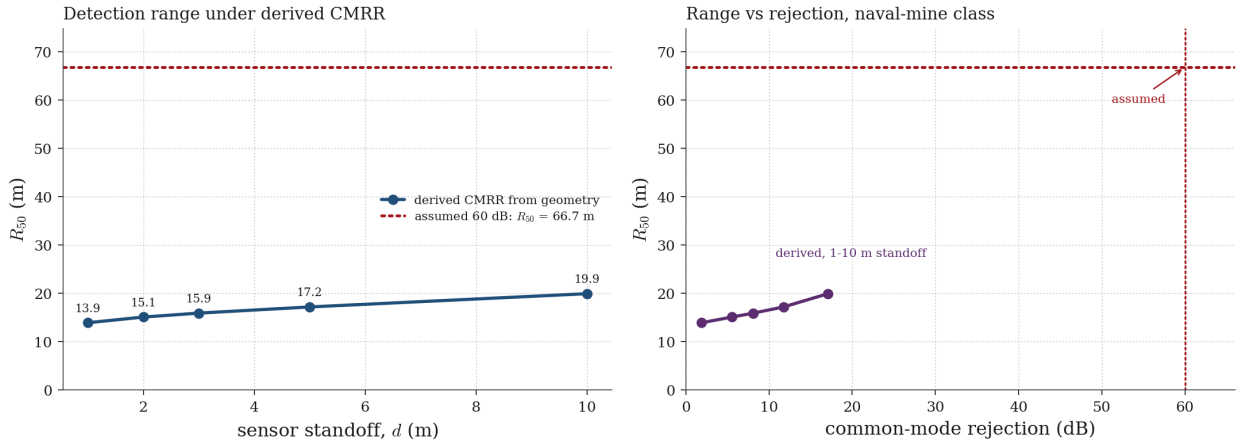


Figure 4: Detection range under the derived rejection[S], against the assumed-rejection baseline of 66.75 m.

cannot deliver that; neither can a tow.

What displacement *does* deliver is a reduction in the absolute clutter field, eightfold per doubling. That is precisely why R-13 remains sound advice and why the Girona 500's 2–3 m standoff [6] is well judged. But it means the two must not be conflated: **standoff reduces how much clutter there is; it does not much improve the gradiometer's ability to reject what remains.**

The design conclusion is that platform interference must be attacked at the source, in absolute terms, by material selection and current-loop control at build time — which is the subject of Section 4 and the thesis of this report. Gradiometry and compensation are second lines of defence, not substitutes.

6 Integration Architecture and TRL Path

6.1 Architecture recommendations

Req	Subsystem	Recommendation	Basis
I-01	Sensor mounting	Rigid boom, not tow cable; baseline held by a structure that cannot flex between cells	[7]; R-09 matching
I-02	Standoff	2–3 m forward of the hull, away from thrusters; treat as clutter reduction, not rejection	[6]; Eq. (3)
I-03	Fasteners	Nitronic 50 (XM-19); do not accept 316L on a non-magnetic certification alone	[15, 16]; Table 2
I-04	Payload structure	CFRP where non-load-bearing; Ti-6Al-4V where load-bearing	Table 2
I-05	Shield	Mu-metal annealed *in final form*; diameter is a first-order parameter	Table 3; [9]
I-06	Shield handling	Treat post-anneal deformation as a functional failure; no field re-forming	[9]
I-07	Cell insulation	Silica aerogel; electrically insulating, so it adds no Johnson noise	[18]; R-07
I-08	Near-cell metals	Avoid high-conductivity alloys (Ni-Al bronze) adjacent to the cell	[18]
I-09	Rejection budget	Do not budget 60 dB against platform sources; use the derived value for the actual geometry	Table 4

Table 5: Integration recommendations. I-09 supersedes the use of a uniform-field rejection figure for near-field platform clutter.

6.2 TRL position and gating experiments

The parent study places the integrated concept at TRL 3–4 with integration risk concentrated in the nulling servo. This work does not advance that position — it sharpens where the remaining risk sits, and adds a second concentration of risk in whole-vehicle magnetic cleanliness. Both are testable cheaply relative to development, and the ordering below is by information gained per unit cost.

1. **Measure M_r of worked fasteners.** The single highest-value measurement in this report. Table 2 is driven entirely by assumed remanence values; a bench measurement of representative 316L and Nitronic 50 fasteners, before and after heading and thread rolling, would replace the largest assumption in the model with data. Cost: a fluxgate, a turntable, and a box of bolts.
2. **Measure formed-shield permeability.** Verify the [9] forming penalty on the actual shield geometry and forming process proposed, since Table 3 turns on it.
3. **Bench-test rigid boom versus tow cable** for gradiometer baseline stability, closing the gap identified in Section 3.2.
4. **Measure the candidate vehicle's magnetic signature** versus distance, replacing the single-anchor calibration of Section 5.2 with a component-resolved map. The parent study's Recommendation 7 already asks for this; the present model makes its output directly consumable.
5. **Implement the eddy-current and Johnson-noise term** so the conductivity column of the schema drives a result, as discussed in Section 8.

7 What Changed and What Did Not

Because this report revises a quantity used in a published study, it is worth being explicit about the boundaries of the claim.

Unchanged. The CAPE requirement of Eq. (1) and the 94 dB figure; the clutter-limited finding; the sensor noise floor and its five-order margin; all twenty-seven verification, validation and stress results; every published figure, which remains byte-identical. Nothing in the parent study has been recomputed or restated incorrectly.

Refined. The interpretation of the 60 dB rejection figure. It remains correct for uniform fields, which is the regime R-09's matching specification addresses. What this work shows is that it does not transfer to near-field platform sources, and that the parent study's own X-07 discussion — which notes that vehicle sources "lie metres from the baseline, their field has a genuine gradient across it and cancellation is necessarily imperfect" — anticipates exactly this and is here made quantitative.

New. A material property layer that did not previously exist; a geometry-derived rejection ratio; the forming-penalty result for shield material; and the remanence-versus-susceptibility distinction as a build-time constraint.

Not claimed. That the parent study's detection ranges are wrong. They are correct given their stated assumption. The derived ranges in Table 4 rest on a vehicle model whose component split is assumed and whose calibration has a single anchor, and they should be read as showing the *sensitivity* of the published figures to a parameter that was previously free — not as a replacement capability estimate.

8 Limitations

1. 84% **of schema property values are assumptions.** The most consequential is the remanence of cold-worked 316L, which drives Table 2 and has no measured basis. The ranking of materials is robust because it follows from whether a ferromagnetic phase can form; the magnitudes are not.
2. **Single-anchor calibration.** The vehicle model matches one measured number [5]. A model with a correct total and an incorrect spatial distribution would produce a plausible common-mode field and an incorrect gradient, and nothing here excludes that.
3. **No eddy-current or Johnson-noise term.** Conductivity is carried in the schema and reported but drives no result. The parent study cites [18] for a shield noise floor that is likewise asserted rather than computed; closing this remains open in both.

4. **Point-dipole sources.** Valid beyond roughly one body length of each source; comfortable at a 2 m standoff from a 0.1 m motor, marginal for the pressure housing treated as a lumped source at the vehicle centre.
5. **Single-layer shielding only.** Eq. (2) does not cover the multi-shell configurations [11] that a real payload would likely use, and which would ease the diameter constraint of Table 3.
6. **Anisotropy discarded.** CFRP conductivity spans roughly seven orders of magnitude between in-plane and through-thickness; a scalar is a poor description and is retained only because no conductivity-driven physics consumes it yet.
7. **No hydrodynamic simulation.** Section 3 is literature only, by scope. The two computed rows in Table 1 use measured attitude figures as inputs to an existing validated model; they are not a hydrodynamic prediction.

9 Conclusion

The parent study concluded that a ruggedised SERF MEMS magnetometer is a physically credible AUV MAD payload whose feasibility is governed by ambient field control and platform magnetic cleanliness rather than by sensor sensitivity. This report accepts that conclusion and examines the second of those two constraints in the terms it actually presents itself: structure, materials, and where things are placed.

Three findings result. Gradiometric rejection of platform sources is limited by geometry rather than by channel matching, giving 5.5 dB at a realistic 2 m standoff against the 60 dB appropriate to a uniform field, and displacement cannot close that gap because rejection improves only linearly in standoff. Magnetic cleanliness is dominated by remanence, which survives ambient-field nulling and is therefore fixed at build time, making fastener metallurgy a first-order design parameter rather than a procurement detail. And the passive shield closes the 94 dB budget only when annealed in its final form, since forming without re-annealing costs an order of magnitude of permeability that cannot be recovered on an assembled payload.

The unifying conclusion is that platform interference must be attacked at the source. Gradiometry, compensation and field control are all second lines of defence, and none of them substitutes for a vehicle that is quiet to begin with. The highest-value next step is inexpensive: measure the remanence of worked fasteners, and replace the largest assumption in this report with a number.

References

[1] *Feasibility of a Ruggedised SERF MEMS Vapour-Cell Magnetometer for Magnetic Anomaly Detection from an Autonomous Underwater Vehicle*, Scientific Report, companion document (2026). Source of the CAPE requirement, the clutter-limited finding, and the verification, validation and stress suites referenced throughout.

[2] J. C. Allred, R. N. Lyman, T. W. Kornack, and M. V. Romalis, "High-sensitivity atomic magnetometer unaffected by spin-exchange relaxation," *Phys. Rev. Lett.* **89**(13), 130801 (2002). DOI:10.1103/PhysRevLett.89.130801.

[3] W. C. Griffith, S. Knappe, and J. Kitching, "Femtotesla atomic magnetometry in a microfabricated vapor cell," *Opt. Express* **18**(26), 27167–27172 (2010). DOI:10.1364/OE.18.027167.

[4] A. Borna, J. Iivanainen, T. R. Carter, J. McKay, S. Taulu, J. Stephen, and P. D. D. Schwindt, "Cross-axis projection error in optically pumped magnetometers and its implication for magnetoencephalography systems," *NeuroImage* **247**, 118818 (2022). DOI:10.1016/j.neuroimage.2021.118818.

[5] M. Oehler *et al.*, "Processing of unmanned underwater vehicle vector magnetometer data," *Geophysical Prospecting* **72** (2024). DOI:10.1111/1365-2478.13556. Source of the 246.90 nT raw and 2.23 nT post-compensation UUV interference benchmark.

[6] M. Seidel, T. Frey, and J. Greinert, "Underwater UXO detection using magnetometry on hovering AUVs," *Journal of Field Robotics* **40**, 848–861 (2023). DOI:10.1002/rob.22159.

[7] CONMAR project / GEOMAR Helmholtz Centre for Ocean Research Kiel, "AUV-based Magnetometry," project product sheet (accessed 2026). Girona 500 with four SENSYS FGM3D/100 UW II three-axis fluxgates at 200 Hz;

states that the rigid construction holding the magnetometers suppresses spatial errors that towed systems usually produce.

[8] "An Affordable Wave Glider-Based Magnetometry System for Marine Magnetic Measurement," *J. Mar. Sci. Eng.* **13**(11), 2089 (2025). DOI:10.3390/jmse13112089. Twin-body towed configuration with an S-shaped cable; measured towed-body pitch within $\pm 5^\circ$ and roll within $\pm 1^\circ$ at 0.56 m/s.

[9] C. Gohil, P. N. Burrows, N. Blaskovic Kraljevic, D. Schulte, and B. Heilig, "Measurements of sub-nT dynamic magnetic field shielding with soft iron and mu-metal for use in linear colliders," arXiv:2009.01519v2 (2020). Measured $\mu_i = 55{,}955 \pm 7$ for annealed mu-metal against 3,670, 3,602 and 4,660 for formed, un-re-annealed foil shields.

[10] D. Dubbers, "Simple formula for multiple mu-metal shields," *Nucl. Instrum. Meth. A* **243**, 511–517 (1986). Source of the single-layer shielding factor of Eq. (2).

[11] T. J. Sumner, J. M. Pendlebury, and K. Smith, "Convectional magnetic shielding," *J. Phys. D: Appl. Phys.* **20**, 1095 (1987).

[12] J. F. Hoburg, "Principles of quasistatic magnetic shielding with cylindrical and spherical shields," *IEEE Trans. Electromagn. Compat.* **38**, 93–103 (1996).

[13] S. Celozzi, R. Araneo, and G. Lovat, *Electromagnetic Shielding*, John Wiley & Sons (2008).

[14] Australian Stainless Steel Development Association, Technical FAQ 3, "Magnetic Effects of Stainless Steels." Annealed 316/317L DC permeability 1.003–1.005 at 200 oersteds.

[15] British Stainless Steel Association, "Effect of cold work and heat treatment on the magnetic permeability of austenitic stainless steels." Grades of high austenite stability retain low permeability in both annealed and cold-worked conditions.

[16] Electralloy / G.O. Carlson, NITRONIC 50 (XM-19, UNS S20910) product data sheet. Typical annealed magnetic permeability 1.004 at 50, 100 and 200 oersteds.

[17] MatWeb material data sheet, "Titanium Ti-6Al-4V (Grade 5), Annealed." Density, tensile and thermal properties.

[18] S.-K. Lee and M. V. Romalis, "Calculation of magnetic field noise from high-permeability magnetic shields and conducting objects with simple geometry," *J. Appl. Phys.* **103**(8), 084904 (2008).

[19] J. Fang, R. Li, L. Duan, Y. Chen, and W. Quan, "Study of the operation temperature in the spin-exchange relaxation free magnetometer," *Rev. Sci. Instrum.* **86**, 073116 (2015).

[20] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley, New York (1998), Ch. 5.